

Homework 09: NumPY and Matplotlib

EEC 3150

For each Python exercise, write a separate Python script. Submit all scripts on Blackboard under the appropriate assignment. Name your script files with the following format:

HW<assignment number>_Ex<exercise
number>_<FirstInitial><LastName>.py

Example: HW01_Ex01_GWest.py

For short answer or math exercises, either do your work on a sheet of paper and upload a picture of it to Blackboard or submit some kind of text file or Word document to Blackboard. You can place all such exercises in a single document as long as you clearly label each exercise. These files should follow a similar naming convention as the Python scripts.

Exercise 1 (10 points, Python)

1. Write a function `Boundary(n, value)` which will create an $n \times n$ matrix in which the elements on the top, bottom, left, and right boundaries `value` while the inside values are equal to 0.
2. Demonstrate your function by calling it and printing the output

Here is an example: `Boundary(4, 1)` should produce:

```
[[1 1 1 1]
 [1 0 0 1]
 [1 0 0 1]
 [1 1 1 1]]
```

Exercise 2 (10 points, Python)

1. Write a function `Truncate(arr)` which an array of any shape and return a new array which is a copy of the old array with the following modifications:
 - Set all values < 0 to be $= 0$
 - Set all values > 0 to be $= 1$
2. Demonstrate your function by calling it and printing the output

Here is an example:

`Truncate(np.array([-1, 2, 10, -4]))`
should produce:

```
[0 1 1 0]
```

Exercise 3 (20 points, Python)

The exercise will have you generate random data, plot it, and index it.

1. Generate random data
 - Create 1D `x`, `y` arrays of 1000 random numbers between -2 and 2
2. Plot the data
 - Plot the `x`, `y` arrays as blue dots

3. Index the data

- Through whatever means you choose, create `x2`, `y2` arrays which only have the points where $-1.5 < x < 1$ and $-1 < y < 0.9$

4. Plot the new data

- Plot the `x2`, `y2` arrays on the same plot as red x's with an increased `markersize` for readability

